Phase 7: Project Documentation

Setup & Installation

1. Clone the repository:

```
git clone https://github.com/Srilatha-dot/Smart_Lender.git
```

2. Navigate into the Project Development Phase folder:

```
cd "Smart_Lender/5. Project Development Phase"
```

3. (Recommended) Create and activate a virtual environment:

```
python -m venv venv
venv\Scripts\activate      (Windows)
source venv/bin/activate    (Linux / macOS)
```

4. Install the required dependencies:

```
pip install -r requirements.txt
```

Running the Application

1. (If model files are not already present) Train the model:

```
cd src
python train.py
```

This loads and cleans the dataset, trains and compares Logistic Regression, Random Forest, and XGBoost, and saves the best-performing model and scaler into the src/models/ folder.

2. Start the Flask web application:

```
python app.py
```

3. Open a browser and go to:

```
http://127.0.0.1:5000/
```

User Guidelines

- Enter the applicant's details in the form: number of dependents, education, employment status, annual income, requested loan amount, loan term, CIBIL score, and asset values (residential, commercial, luxury, and bank assets).
- Click the "Predict" button to submit the details to the backend.

- The result panel instantly displays whether the loan is Approved or Rejected, along with the approval probability, without reloading the page.
- All numeric fields should be entered as positive numbers; the CIBIL score should be entered in the standard 300–900 range for a realistic prediction.
- If an error message is shown instead of a result, check that all required fields have been filled in correctly.

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